

Telco Customer Churn Analysis Report

Introduction

Customer churn is one of the most important challenges faced by telecommunication companies. Customer churn occurs when a customer stops using a company's services and switches to a competitor or discontinues the service entirely.

The purpose of this analysis is to identify the key factors contributing to customer churn and provide recommendations that can help improve customer retention and reduce customer loss.

Objective

The main objectives of this project are:

- Analyze customer churn behavior.
 - Identify factors influencing customer churn.
 - Examine the relationship between contract type, payment method, tenure, and churn.
 - Provide actionable recommendations to improve customer retention.
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Dataset Overview

The dataset contains customer information including:

- Customer ID
- Gender
- Senior Citizen Status
- Partner and Dependents
- Tenure
- Phone Service
- Internet Service
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn Status

The target variable used for analysis is:

Churn

- Yes = Customer left the company
 - No = Customer remained with the company
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Data Cleaning and Preparation

The following preprocessing steps were performed:

- Checked for missing values.
- Converted Total Charges column to numeric format.
- Replaced blank values with 0 where appropriate.
- Verified duplicate records.
- Converted Senior Citizen values from 0 and 1 into Yes and No for easier interpretation.
- Validated data types and column consistency.

These steps ensured the dataset was clean and suitable for analysis.

Exploratory Data Analysis

Several visualizations were created to understand customer behavior and identify churn patterns.

The analysis focused on:

- Churn distribution
 - Contract type
 - Payment method
 - Customer tenure
 - Senior citizen status
 - Internet service type
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Key Findings

Overall Churn Rate

The analysis revealed that approximately 27% of customers have churned, while approximately 73% of customers remained with the company.

This indicates that although the majority of customers stay with the company, a significant portion is still being lost.

Contract Type Analysis

Contract type emerged as one of the strongest predictors of customer churn.

Churn Rate by Contract Type

Contract Type	Approximate Churn Rate
Month-to-Month	42%
One Year	11%
Two Year	3%

Insight

Customers on month-to-month contracts are significantly more likely to leave compared to customers on long-term contracts.

Long-term contracts create stronger customer commitment and improve retention.

Payment Method Analysis

Payment method showed a strong relationship with customer churn.

Churn Rate by Payment Method

Payment Method	Approximate Churn Rate
Electronic Check	45%
Credit Card	15%
Bank Transfer	17%
Mailed Check	19%

Insight

Customers using electronic checks show the highest churn rate.

This may indicate lower customer satisfaction, reduced convenience, or trust-related concerns associated with this payment method.

Customer Tenure Analysis

Customer tenure plays a major role in retention.

Churn Rate by Tenure

Customer Tenure	Approximate Churn Rate
Less than 1 Year	50%
1 – 3 Years	35%
More than 3 Years	15%

Insight

Customers are most likely to churn during their first year.

As tenure increases, churn decreases significantly.

This suggests that early customer engagement is critical for long-term retention.

Internet Service Analysis

Churn Rate by Internet Service

Service Type	Approximate Churn Rate
Fiber Optic	30%
DSL	20%

Insight

Customers using Fiber Optic services exhibit higher churn rates.

This may indicate service quality concerns, pricing issues, or stronger market competition within this segment.

Senior Citizen Analysis

Churn Rate by Age Group

Customer Group	Approximate Churn Rate
Senior Citizens	41%
Non-Senior Citizens	26%

Insight

Senior citizens are more likely to churn compared to younger customers.

Additional customer support and tailored retention programs may help improve satisfaction within this group.

Visual Analysis

The dashboard visualizations provided clear evidence of churn patterns:

- Bar charts highlighted churn differences across payment methods and contract types.
- Pie charts displayed overall churn distribution.
- Line charts showed the relationship between tenure and churn.
- Comparative visualizations revealed high-risk customer groups.

These visualizations made it easier to identify the factors most strongly associated with customer churn.

Recommendations

Based on the analysis, the following recommendations are suggested:

1. Promote Long-Term Contracts

- Offer discounts for annual and multi-year subscriptions.
- Provide loyalty rewards for contract renewals.

Expected Outcome:

- Reduced churn rates.
 - Improved customer retention.
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2. Improve Payment Experience

- Encourage customers to switch from electronic checks to automated payment methods.
- Offer incentives for using credit cards or bank transfers.

Expected Outcome:

- Increased customer convenience.
 - Lower churn rates.
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3. Focus on New Customers

- Implement onboarding programs during the first year.
- Conduct regular follow-ups with newly acquired customers.

Expected Outcome:

- Higher early-stage engagement.
 - Improved long-term retention.
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4. Improve Fiber Optic Customer Satisfaction

- Investigate service quality concerns.
- Review pricing strategies.
- Strengthen customer support.

Expected Outcome:

- Reduced churn among Fiber Optic users.
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5. Develop Senior Citizen Retention Programs

- Provide dedicated support channels.
- Offer personalized plans and discounts.

Expected Outcome:

- Increased satisfaction and loyalty among senior customers.
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Conclusion

The analysis identified contract type, payment method, tenure, internet service, and senior citizen status as the primary factors influencing customer churn.

Customers on month-to-month contracts, customers using electronic checks, customers with short tenure, Fiber Optic users, and senior citizens showed significantly higher churn rates.

By implementing targeted retention strategies and improving customer engagement, the company can reduce churn, improve customer loyalty, and increase long-term revenue.